IBEHS 5P06: Engineering Design Quality – Graded Instructor Meetings

Weight: 20% of the final course grade, Eight meetings at 2.5% each
Schedule: Four graded meetings (two per term) with the BME instructor and four graded meetings (two per term) with the disciplinary instructor(s)

The graded instructor meetings evaluate the quality of the team's engineering design process across the full year. They provide recurring checkpoints for students to demonstrate preparation, technical progress, engineering judgement, responsiveness to evidence and feedback, and a credible plan for subsequent work.

Meeting requirements

- Team responsibility: The team is responsible for initiating, planning, and scheduling each graded meeting according to the process established by the relevant instructor. Instructors are not responsible for reminding teams to arrange their meetings.
- Scheduling: Contact the instructor and teaching assistant sufficiently in advance to arrange a meeting time. Unless otherwise specified by the instructor, the meeting should be scheduled at least one week in advance.
- Agenda and preparation: The team must prepare and send a proposed agenda to the instructor at least one week before the meeting. The agenda should identify the topics to be discussed, decisions or feedback required, and any evidence or materials the instructor should review.
- Meeting materials: Any supporting materials needed for a productive discussion should be organized and available before the meeting. Teams should not use meeting time to locate, assemble, or interpret basic project information.
- Attendance: All team members are expected to attend the full graded meeting unless an absence has been approved in advance according to course policies.
- Participation: All team members must be prepared to discuss the overall project, relevant design decisions, current progress, and next steps. Each student must also be able to explain their individual contributions and discipline-specific technical work.
- Meeting management: The team is responsible for leading the meeting, managing the agenda and time, recording key feedback and decisions, and identifying any resulting action items. The team is responsible for initiating and scheduling each graded meeting according to the process established by the relevant instructor. Students should not rely on the instructor to initiate or remind them about the meeting.

1. Meeting structure

- Each team will complete four graded meetings with the BME instructor and four graded meetings with the appropriate disciplinary instructor(s).
- Each meeting is worth 2.5% of the final course grade, for a total of 20%.
- For teams involving two disciplines, the discipline instructor's grade will apply only to students from that discipline. However, all team members must attend the meetings, even though only students from the relevant discipline will be assessed.
- The agenda, preparation requirements, evidence, and discussion format will be determined at least one week before the meeting with the relevant instructor and teaching assistant to reflect the project stage and disciplinary context.
- In addition to the graded meetings, students are expected to meet regularly with their instructors according to a schedule agreed upon with the instructors.
- Meeting length and format: Graded meetings should be approximately 20–30 minutes, unless otherwise arranged with the instructor. Meetings are intended to be working discussions rather than formal presentations. Teams should not prepare a PowerPoint presentation unless specifically requested by the instructor. Relevant project materials, figures, calculations, models, prototypes, or other evidence may be shown informally on a laptop or in another appropriate format to support the discussion.

2. **Student responsibilities**
   - All team members must attend each graded meeting and participate meaningfully.
   - Provide the instructor with enough current information to understand the project status without reconstructing work from incomplete records.
   - Present evidence of progress, including relevant design records, analysis, prototypes, verification work, stakeholder input, risk management, and project-planning updates.
   - Meet with stakeholders 1 to 2 times a month and use the resulting evidence and feedback to improve the design where appropriate.
   - Be prepared to explain decisions, assumptions, limitations, constraints, setbacks, changes, and the rationale for next steps.
3. **Evidence of engineering design quality**
   - Demonstrate how the aspects of the design process most relevant to the team's current stage of work are connected. This may include user needs, design inputs, design outputs, risks, standards, analysis, implementation, and verification. Teams are not expected to address every element in every meeting, but should be prepared to explain how the topics discussed fit within the broader design process.
   - Show appropriate discipline-specific technical work and engineering judgement.
   - Demonstrate meaningful iteration and appropriate responses to stakeholder, instructor, TA, and experimental feedback.
   - Maintain realistic responsibilities, resources, timelines, contingency plans, and progression metrics.
4. **Grading and individual adjustment**
   - The default result for each meeting is a group grade.
   - The instructor may adjust an individual student's grade when attendance, preparation, participation, technical contribution, knowledge, or demonstrated responsibility differs materially from the team's overall performance.
   - Because all students must attend, absences must be handled through the course's accommodation and rescheduling processes.
5. **Relationship to other assessments**
   - Progress Reports are formative Pass/Fail assessments graded by teaching assistants and are separate from Engineering Design Quality.
   - Students are expected to use progress-report feedback to improve the design process and the evidence presented in later instructor meetings and the Final Project Report.
   - The DHF should provide a current and traceable record supporting the team's claims during each meeting.

6. **Graded Meeting Deadlines**
   - The following dates are the last days that you can have your graded meetings:
     - First meeting: Oct 31
     - Second Meeting: Nov 30
     - Third Meeting: Feb 12
     - Fourth Meeting: March 18

Meeting-readiness checklist

- The agenda and required pre-meeting materials have been agreed upon with the instructor and sent prior to the meeting.
- The DHF, Gantt chart, and relevant technical evidence are current and accessible.
- The team can explain progress since the previous meeting, important decisions, unresolved risks, and the next stage of work.
- Every student is prepared to participate and answer questions.